Belief-State Control for Diplomatic Negotiation with a Lower-Cost Language Model

Anonymous ACL submission

Abstract

Can explicit strategic control improve negotiation with a lower-cost language model? We study a simulation grounded in 12 real diplomatic documents, holding GPT-5.6 Luna's generated utterances and their interpretations fixed across controllers. On 64 matched conditions, belief-state planning achieves net utility 0.533, compared with 0.514 for direct action selection by GPT-5.6 Sol with medium reasoning. Planner-distilled PPO reaches 0.475 without further model calls for strategy selection. Separate interventions attribute most of the learned policy's gain to signals and updated beliefs. Dialogue analysis connects these mechanisms to conditional offers and commitment wording. The results identify the benefits of explicit control under declared negotiation rules.

1 Introduction

Large language model (LLM) agents create opportunities to negotiate on behalf of users (Yao et al., 2023; Bianchi et al., 2024). They must pursue user goals while responding to parties with different interests and incomplete information. Existing benchmarks show that strategic choices affect outcomes (Abdelnabi et al., 2024). Repeated decisions also make inference cost relevant (Kapoor et al., 2024), motivating systems that assign specialized roles to lower-cost models (Belcak et al., 2025).

Prior work separates strategy from language generation (He et al., 2018), combines dialogue with planning (Meta Fundamental AI Research Diplomacy Team (FAIR), 2022), and improves smaller negotiators through reinforcement learning and distillation (Hua et al., 2026; Tabacof and Joglekar, 2026). Building on these approaches, we ask: with language realization held fixed, how does explicit strategic control affect negotiation utility and the cost of choosing actions?

We construct a diplomatic negotiation simulation from 12 real documents across six event groups. The documents provide context and examples of reciprocal conditions; utilities, hidden opponent types, and responses are synthetic. GPT-5.6 Luna (Luna) generates and separately interprets 396 fixed utterances. We compare Random, Heuristic, Bandit, PPO, direct model decisions, an exact public-belief planner, and a policy distilled from that planner (Figure 1). The planner knows the simulation rules but never observes the hidden type.

On 64 matched conditions, the planner achieves higher utility than direct action selection by GPT-5.6 Sol (Sol) with medium reasoning. Distilled PPO improves on base PPO and acts without further model calls for strategy selection. We then use interventions and matched dialogue traces to explain these results. This analysis distinguishes questions that reveal information from signals that change acceptance incentives, and examines whether actions preserve the conditions and commitments expressed in dialogue. Our contribution is a controlled comparison linking negotiation outcomes, strategic computation, and discourse meaning.

2 Related Work

Negotiation agents. Lewis et al. (2017) learn from human bargaining dialogues and improve decisions through simulated continuations. Recent benchmarks cover resource allocation, trading, and multiple stakeholders (Bianchi et al., 2024; Abdelnabi et al., 2024). CICERO combines dialogue and strategic reasoning in the board game Diplomacy (Meta Fundamental AI Research Diplomacy Team (FAIR), 2022). Our simulation uses real diplomatic documents to frame reciprocal measures and commitments.

Strategic control. He et al. (2018) separate dialogue acts from linguistic realization. GDP-Zero uses LLM simulations for tree search (Yu et al., 2023); PPDPP learns a separate policy planner

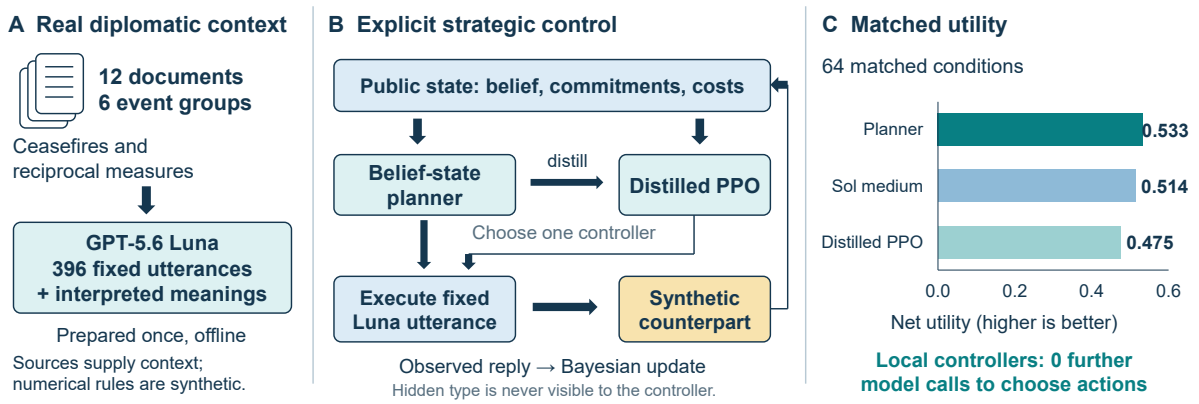


Figure 1: Overview. (A) Real diplomatic sources provide context for a fixed set of Luna utterances and interpreted meanings. (B) A public-belief planner or its distilled PPO policy selects actions; observed responses update the public state. Distillation transfers planner supervision to the policy, while language stays fixed. (C) Matched utility with the same language execution. Zero further model calls refers only to local strategy selection.

(Deng et al., 2024); and Dual-Mind combines strategy planning with expression refinement (Liu et al., 2025). We build on this separation by holding utterances and interpretations fixed when comparing controllers, then analyzing their information use and commitments.

Efficient agents. Kapoor et al. (2024) advocate evaluating performance alongside cost, while Belcak et al. (2025) argue for specialized small-model roles. SocialRL combines reinforcement learning and teacher distillation for negotiation (Hua et al., 2026). Tabacof and Joglekar (2026) study reinforcement learning across seller model sizes. We keep language model weights fixed and train a separate controller, measuring strategy-selection calls and examining the mechanisms behind utility gains.

3 Method

Data and task. We use 12 document contexts from six diplomatic event groups, including CMFA PressCon (Mochtak and Turcsanyi, 2021) and Security Council discourse. Training covers the Korean Peninsula, South Sudan, and Kashmir (six contexts); development uses Gaza (two); testing uses Syria and Iran (four). Passages about ceasefires, reciprocal measures, and commitments guide synthetic bilateral negotiations. One party offers to maintain its measure for $x \in \{1, 2, 3\}$ weeks if the counterpart maintains its measure for $y \in \{1, 2\}$ weeks. Durations, payoffs, types, and responses are researcher-defined.

State, actions, and reward. The agent has at most five decisions and cannot observe the counterpart's type, $\tau \in \{S, H\}$. It observes the offer, acceptance, commitments, costs, and belief $b_t = P(\tau = H)$. Eleven actions cover ambiguity, signaling, a private question, six conditional offers, ratification, and withdrawal. Questions provide noisy type cues; offer responses also reveal information. Ratification requires observed acceptance and own utility at least as high as the outside option.

Specifically, an agreement gives $u_A = 0.5y - 0.18x$, and the counterpart accepts when $u_B = 0.35x - c_\tau y + \delta z \geq 0$, where z records a signal and c_τ determines the cost of reciprocal measures. A signal therefore changes an incentive; a question changes information. For response likelihoods L_H and L_S, the public belief becomes

$$b_{t+1} = \frac{b_t L_H}{b_t L_H + (1 - b_t) L_S}. \quad (1)$$

Equal response likelihoods leave the belief unchanged. Step rewards are

$$r_t = d_t u_{\text{end}} - 0.01 - c_t^{\text{act}} - 0.005 e_{t+1}^2, \quad (2)$$

where d_t marks termination, e counts rejected offers, and c_t^{act} sums question, signal, rejection, reversal, and invalid-action charges. Terminal utility is u_A at agreement or 0.12 at withdrawal or expiration, paid once. We report $\sum_t r_t$. Appendix A gives all costs and transition rules.

Shared language execution. GPT-5.6 Luna generates three utterance styles for each context–action pair, yielding 396 utterances. A separate Luna call interprets each utterance's function, reciprocal terms, and stated cost. These interpreted meanings

determine transitions, and mismatches with the intended action are retained. Every controller uses this same fixed set. Counterpart replies use fixed templates. This separates action selection from language realization while keeping the consequences of interpreted utterances in the environment.

Strategic controllers. We compare Random, a development-selected signal-first Heuristic, an immediate-reward contextual Bandit, and PPO (Schulman et al., 2017). An exact belief-state planner maximizes expected net utility using the declared response rules and interpreted utterances, without access to the true type. To obtain a learned controller, we distill its action values and then refine the policy with PPO. Teacher targets come only from training contexts and configurations. The distilled policy acts without invoking the planner. We also test direct action selection by Luna with reasoning set to none, and GPT-5.6 Sol with none or `medium`. Direct models receive the public state and numerical rules.

The planner evaluates future response branches by backward recursion over the remaining decisions. Distillation uses 43,440 public states from planner trajectories and exploratory trajectories. The policy learns a soft distribution over the eleven action values, with temperature 0.03, together with a value estimate. After 2,400 supervised updates, PPO refines the same network through interaction. At deployment, it receives 31 public features and chooses its highest-scoring action. Thus, planning supplies offline supervision, while the learned policy replaces repeated planning with a local network evaluation.

4 Experimental Setup

The initial evaluation crosses four test contexts with 64 numerical configurations and three styles. A matched comparison uses the first 16 configurations and one style, giving 64 context–configuration conditions. A follow-up uses 64 new configurations with the same test contexts and three styles. We compute expectations over hidden types and question responses, then average within event groups and across groups. Learned results average three seeds. Bandit and base PPO each receive 102,400 training steps per seed; distilled PPO receives additional teacher supervision. Final checkpoints are used, with method selection on development data. Appendices B and D provide settings and diagnostic analyses.

Controller	Utility	Agree.	Calls
A. Initial: 256 conditions, three styles			
Random	-0.146 ± 0.001	11.0	0
Bandit	0.118 ± 0.001	23.8	0
Heuristic	0.419	94.8	0
PPO	0.358 ± 0.018	73.0	0
Planner	0.488	93.3	0
B. Matched: 64 conditions, one style			
Luna none	-0.368	24.3	3.949
Sol none	-0.367	26.3	4.773
Sol medium	0.514	100.0	3.429
Heuristic	0.470	100.0	0
PPO	0.378 ± 0.012	81.9	0
Distilled PPO	0.475 ± 0.016	98.3	0
Planner	0.533	100.0	0

Table 1: Policy comparison with fixed Luna utterances. Agreement is a percentage; calls are per negotiation for strategy selection. ±: three-seed SD (evaluation seeds for Random). Panels use different conditions.

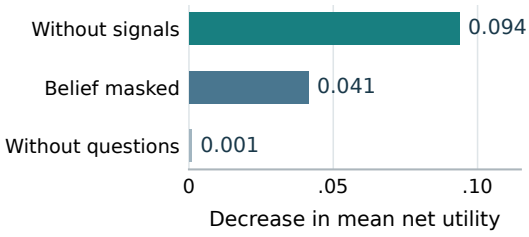


Figure 2: Evaluation-time interventions on distilled PPO. Losses are differences in three-seed mean utility on the follow-up conditions. Full values and seed variation appear in Table 4.

5 Results

Table 1 separates the initial and matched evaluations. In the initial evaluation, PPO outperforms Random and Bandit but falls below the heuristic. In the matched comparison, the planner reaches 0.533 utility, compared with 0.514 for Sol medium. Distilled PPO reaches 0.475, improving on base PPO (0.378) and slightly exceeding the heuristic (0.470). Sol none remains near Luna none, whereas Sol medium performs substantially better. Thus, the direct-model results depend strongly on the tested reasoning setting.

Local controllers require no language model calls for action selection. Sol medium uses an expected 3.429 calls per negotiation, corresponding to 4,184.8 tokens and $0.02363 at the recorded request costs. These figures count logical requests even when experimental caching avoids a repeated call. These strategy-selection figures exclude shared utterance construction, interpretation,



Figure 3: Selected paths under the same Iran-context condition and soft counterpart. Terms $x{:}y$ denote own and reciprocal durations. Values are episode net utility; the learned path uses seed 17.

and offline training.

On the follow-up configurations, distilled PPO achieves 0.468 utility, compared with 0.364 for base PPO and 0.445 for the heuristic (Figure 2). Blocking signals reduces utility by 0.094; replacing the posterior feature with the prior reduces it by 0.041. Blocking questions reduces it by only 0.001. These are evaluation-time interventions without retraining. The learned policy benefits mainly from signals and updated beliefs in this setting. Updated beliefs can come from offer responses as well as questions. The planner's no-question comparison yields a larger utility loss of 0.008.

Discourse analysis. Matched Iran-context traces illustrate this distinction (Figure 3). Luna none repeats a rejected 1:2 offer four times and withdraws, yielding -0.80. The planner and Sol medium ask a private question, send a signal, and obtain acceptance of that same offer, yielding 0.70. The question changes the belief from 0.65 to 0; the signal raises the soft counterpart's acceptance utility from -0.05 to 0.05. The learned controller instead signals and offers 2:2, yielding 0.55. Wording also matters: in a Syria-context pair, explicit withdrawal terminates the episode, while conditional withdrawal is interpreted as invalid and incurs repeated costs. Appendix C examines these paths and distinguishes expressed readiness to ratify from ratification itself.

These paths show why action order matters. The repeated 1:2 demand is rejected by both types before signaling, so rejection provides no new type information. A question can resolve uncertainty, but it does not itself make the demand acceptable. The subsequent signal changes that condition. The learned policy reaches agreement by offering a larger own concession instead. Its successful agreement therefore does not imply that it reproduces the planner's information-gathering strategy. Comparing speech-act sequences reveals this difference even when both paths end in ratification.

6 Discussion

Our results support assigning explicit strategic decisions to a separate controller when the negotiation rules can be specified. The planner uses beliefs, action costs, and future consequences to improve decisions under the same language conditions. Distillation transfers part of this behavior to a policy that acts without search or further language model calls. Its improvement over base PPO shows the value of planner supervision in this setting. Its remaining gap to Sol medium and small advantage over the heuristic distinguish exact planning from learned control.

The interventions suggest that asking more questions is not the main source of the learned policy's improvement. Signals change acceptance incentives, while responses to offers can update beliefs without a separate question. Useful information therefore depends on what the agent can subsequently do with it and what obtaining it costs. Discourse analysis adds a complementary requirement: the chosen action must be expressed with the intended conditions and commitment strength. A conditional threat to withdraw and an actual withdrawal can lead to different transitions. Strategic evaluation should therefore connect utility, information use, and the meaning of the resulting utterance.

7 Conclusion

We studied strategic control for a lower-cost language model in a diplomatic negotiation simulation grounded in real documents. With utterances held fixed, belief-state planning achieved the highest utility in the matched comparison, and planner-distilled PPO improved on base PPO without further model calls for action selection. Interventions linked the learned policy's gains mainly to signals and updated beliefs. Dialogue traces further showed how information use and commitment wording shape negotiation paths. These findings identify explicit strategic control as a useful component of negotiation systems and show how controlled comparisons can explain its benefits.

Limitations

Real documents ground the task, but payoffs and dialogues are synthetic. Evaluation covers four contexts from two event groups; extra numerical settings are not independent historical cases. Fixed utterances limit claims about online dialogue. All 44 matched style-0 meanings equal canonical templates, so utility does not establish a generation advantage. Shared-model interpretation lacks independent human validation, and some commitment expressions are misclassified.

The planner knows the response rules and interpretation mapping; distillation adds teacher supervision and computation. These resources limit comparisons with model-free learning. Model mismatch remains untested, and the tested reasoning settings do not isolate model-size effects. Three seeds and one cached model response per state do not capture historical or model-output uncertainty. Cost comparisons cover strategy-selection calls, excluding language preparation, teacher and training costs, and local computation. Detailed assumptions and procedures appear in Appendices A–C.

Ethical Considerations

Public diplomatic statements provide context, not evidence of private preferences or predictions of state behavior. Synthetic payoffs and escalation indices should not guide real diplomatic decisions. Source identifiers and transformations are preserved, and access to the supplied corpus remains restricted to the research collaboration. Language models assisted utterance generation, interpretation, manuscript drafting, and a visual concept; the final teaser was redrawn as editable shapes using recorded results. Model interpretation checks are not human annotation or evidence of independent expert agreement.

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A Environment and Data

Payoffs and response model. The agent's terminal agreement utility and counterpart's acceptance utility are

$$u_A(x, y) = 0.5y - 0.18x, \quad (3)$$

$$u_B(x, y, \tau, z) = 0.35x - c_\tau y + \delta z. \quad (4)$$

Here $c_S = 0.2$, c_H is configuration-specific, and z indicates whether a signal has been sent. The counterpart accepts if $u_B \geq 0$. Ratification additionally requires observed acceptance and $u_A \geq 0.12$. Withdrawal or expiration returns terminal utility 0.12 before accumulated costs. Agreement rates throughout the paper refer to this implemented terminal transition.

For the first private question, $P(o = H \mid H) = q$ and $P(o = H \mid S) = 1 - q$. Later questions cost the same but provide no new cue. Offers receive deterministic acceptance or rejection conditional on type. For likelihoods L_H and L_S of the observed response,

$$b_{t+1} = \frac{b_t L_H}{b_t L_H + (1 - b_t) L_S}. \quad (5)$$

The hidden type generates responses but is never an input to any controller. Beliefs are updated by the environment from interpreted meanings and responses. Equal type likelihoods leave the belief unchanged.

Costs and commitments. Each decision costs 0.01. Questions incur c_q; signals incur the expenditure extracted from their utterance (intended cost 0.06). Sending a signal sets $z = 1$; repeated signals incur cost but do not stack their benefit. Each rejected offer incurs c_r and increments escalation e. Every turn also costs $0.005e_{t+1}^2$. Reducing one's offered duration below the recorded commitment costs 0.08. Ambiguity makes the next offer provisional, preventing that offer from replacing the recorded public commitment; it does not erase an earlier commitment or remove a reversal charge. Invalid interpretations and invalid ratification each cost 0.10. Rewards sum terminal utility, when present, minus these costs. We report undiscounted net utility even when training uses discounted returns.

Public features. The 31-dimensional observation includes time, posterior belief, offer durations, acceptance, signal and question flags, provisional status, commitment, escalation, prior, question reliability, numerical utilities and costs, a three-way style indicator, and a seven-way indicator of the last interpreted function. It excludes source identity and hidden type. Direct-model prompts receive the public state, configuration, last interpreted function, and the numerical decision rules. Each unique prompted state has one saved model response; this is not repeated sampling over model outputs.

Parameter	Candidate values
Prior b_0	.20, .35, .50, .65, .80
Question reliability q	.60, .75, .90, 1.00
Question cost c_q	.01, .02, .04, .08
Rejection cost c_r	0, .04, .08, .16, .32
Signal effect δ	0, .10, .25
Hard-type cost c_H	.40, .45, .55, .90

Table 2: Synthetic parameter values. We draw 96 training, 32 development, and 64 initial test configurations. The follow-up draws 64 configurations excluded from all earlier configuration sets.

Source provenance. The supplied source collection contains diplomatic briefings, public statements, and Security Council discourse. Table 3 identifies every selected context. Full source sentences, URLs or dataset identifiers, and the authored bilateral interpretation are preserved in `study_results/contexts.json`. Documents determine the issues and linguistic context; they do not supply observed utility functions or historical negotiation trajectories. The event groups are disjoint across training, development, and testing. Some source material had been inspected during earlier exploration, so the test split denotes exclusion from policy training rather than researcher-blind historical evidence.

Language checks. The three styles are a formal communiqué, a direct private statement, and a concise statement with reordered clauses. Intended and interpreted meanings match in 379 of 396 entries; all 17 mismatches remain in the environment. In 180 controlled probes covering valid offers, changed durations, negation, unless-conditions, and missing reciprocity, the interpreter matches the authored expectation in all cases. These are automatic consistency checks. They do not establish human-validated speech-act accuracy. Twelve source contexts also have two type-specific validation paths each (24 total).

ID	Split / event group	Source identifier	Date
kr1	train / peninsula	CMFA PressCon dyad 14733	2017-04-05
kr2	train / peninsula	CMFA PressCon dyad 16629	2018-03-28
ss1	train / south sudan	CMFA PressCon dyad 10890	2014-05-12
ss2	train / south sudan	CMFA PressCon dyad 13454	2016-07-12
ss3	train / south sudan	UNSC S/PV.9467 (Albania)	2023-11-06
ip1	train / kashmir	CMFA PressCon dyad 10527	2013-12-18
ga1	dev / gaza	Biden–Kishida joint statement	2024-04-10
ga2	dev / gaza	US State Department press briefing	2024-03-25
sy1	test / syria	CMFA PressCon dyad 12850	2016-03-01
sy2	test / syria	UNSC S/PV.8727 (Turkey)	2020-02-19
ir1	test / iran	UNSC S/2020/644, speech 0014	2020-06-30
ir2	test / iran	CMFA PressCon dyad 10285	2013-10-17

Table 3: Selected source contexts. CMFA denotes the Chinese Ministry of Foreign Affairs briefing corpus in the supplied material. The stored source manifest preserves exact sentences and provenance.

Controller / intervention	Utility	Change
PPO	0.364 ± 0.016	–
Heuristic	0.445	–
Distilled only	0.446 ± 0.011	–
Distilled PPO	0.468 ± 0.012	–
Without questions	0.467 ± 0.014	-0.001
Belief masked	0.427 ± 0.013	-0.041
Without signals	0.374 ± 0.014	-0.094
Planner	0.500	–
Planner without questions	0.492	-0.008

Table 4: Follow-up: 256 context–configuration conditions and three styles, using new numerical configurations on the same test documents. Changes are relative to distilled PPO or, for the final row, the full planner. $\pm$ denotes three-seed SD.

B Training and Additional Results

The heuristic is selected from six fixed candidates using development net utility. The selected policy sends a signal before choosing an offer with high belief-weighted immediate value, ratifies eligible agreements, and withdraws when its rule favors the outside option. Random samples uniformly from 11 actions. The contextual Bandit uses a linear immediate-reward estimator per action with exploration probability 0.2.

Base PPO uses two 64-unit hidden layers, learning rate 3×10^{-4}, discount 0.95, entropy coefficient 0.01, rollout length 256, batch size 64, and ten optimization epochs. Bandit and PPO use training seeds 17, 29, and 43. Random uses evaluation seeds 101, 202, and 303 with four repetitions per response branch and style.

The planner enumerates transitions using the actual interpreted utterance set for each context and style. It solves the finite-horizon problem by backward recursion, averaging response branches under the current public belief. The no-question planner excludes actions interpreted as questions. This planner has exact knowledge of the declared simulation and utterance meanings.

Distillation uses 43,440 public-state examples from training contexts and numerical settings, collected from planner trajectories and trajectories with 35% random exploration. Targets are the planner's eleven action values $Q(a)$ and maximum value. The actor target is $p(a) \propto \exp(Q(a)/0.03)$. Training minimizes cross-entropy plus 0.5 times value mean-squared error, using 2,400 updates of 256 samples, Adam learning rate 0.001, and gradient clipping at 0.5. The same network then receives 102,400 PPO interactions. Development utility averaged over three seeds selects distilled PPO (0.465) over distillation alone (0.427). Teacher computation is additional to the interaction budget.

At evaluation, learned controllers choose their highest-scoring action. Blocking a question or signal excludes its action index and selects the highest remaining logit. Belief masking replaces only the posterior feature by the original prior. Other public observations, including accepted offers, remain visible. These interventions do not remove every possible source of type information and do not retrain the controller.

Aggregation and uncertainty. For each document–configuration condition, we enumerate both latent types and all nonzero first-question response branches with their declared probabilities. We average styles, then configurations and contexts within event groups, then event groups equally. The reported SD quantifies variation across three training seeds, or evaluation seeds for Random. It is not a confidence interval over historical events. Initial and follow-up evaluations each contain 256 document–configuration conditions, not 256

independent diplomatic events. Each matched method has 200 nonzero latent/response paths across its 64 conditions.

Supplementary training interventions. The initial experiment varies the discount (0.8, 0.95, 0.99), entropy coefficient (0, 0.01, 0.1), and inclusion of rejection and commitment-reversal costs during training. All variants are evaluated under the common original reward. Non-base variants use seed 17 and are exploratory. Their net utilities are 0.346 and 0.327 for discounts 0.8 and 0.99; 0.366 and 0.392 for entropy 0 and 0.1; 0.377 without training rejection costs; and 0.371 without training reversal costs. The corresponding seed-17 baseline and complete metrics are preserved in `study_results/seed_metrics.csv`. Reward components, counterpart utility, agreement, escalation, reversals, length, learning curves, and action-transition counts are included in the released results. The initial evaluation's weighted feasible fraction is 95.3%; the follow-up fraction is 97.3% under canonical offer and signal options. No reported terminal agreement violates the numerical bilateral acceptance checks.

C Discourse Cases

We examine six stored episodes: four policies under one Iran-context condition, and a pair of Syria-context episodes with the same policy and numerical condition but different utterance styles. These are synthetic, source-grounded exchanges selected to explain decision paths and interpretation effects. They are not historical transcripts or a blinded human annotation study. The hidden types below describe evaluation branches, not information available to the agent.

Iran source and shared condition. Context `ir1` uses the 30 June 2020 Security Council passage S/2020/644 attributed in the supplied corpus to Halimah DeShong. The passage describes Iran's steps as reversible following US withdrawal from the agreement. Our authored setting uses conditional sanctions suspension and nuclear-activity suspension. In condition 0, the actual evaluation type is soft, $b_0 = 0.65$, $q = 1$, $c_q = 0.02$, $c_r = 0.16$, $\delta = 0.10$, and $c_H = 0.45$; style 0 is fixed across policies.

Episode 1: Luna none. The agent says, "We will suspend the specified sanctions for 1 week, conditional upon your suspension of the specified nuclear activities for 2 weeks; our commitment is contingent upon yours." The offer is rejected. The same text is repeated three more times, followed by withdrawal. Both types reject the unsignaled offer, so every rejection leaves $b = 0.65$. Repetition therefore adds no type information while accumulating rejection and escalation costs. Terminal outside utility 0.12 minus total costs 0.92 yields net utility -0.80.

Episode 2: Belief-state planner. The planner asks, "Privately, we ask whether you require a larger reciprocal concession before accepting this understanding. This inquiry is exploratory only and constitutes neither an offer nor a public commitment." A soft response changes b to zero. The agent then commits resources at stated cost 0.06 to a monitored confidence-building measure. The same 1:2 offer is now accepted: the signal changes soft-type utility from $0.35 - 0.2(2) = -0.05$ to 0.05. The question identifies the type, while the signal changes the acceptance incentive. Agreement utility 0.82 minus four turn costs, question cost, and signal cost yields 0.70.

Episode 3: Sol medium. Sol selects the same question–signal–offer–ratification path and receives 0.70. Since all methods use the shared utterances, identical actions produce identical text here. This trace shows agreement between two controllers under the same state and language conditions; it is not a second independently generated dialogue style.

Episode 4: Distilled PPO, seed 17. The learned policy sends the signal without asking a question, then offers to suspend sanctions for two weeks if the counterpart suspends nuclear activities for the same period. Acceptance identifies the soft type in this condition, reducing b from 0.65 to zero. The policy then selects ratification. Its larger concession reduces terminal utility to 0.64. Three turn costs and signal expenditure leave net utility 0.55. This path obtains information through an offer, but pays for it through less favorable terms.

Commitment strength in Episodes 2–4. The final shared utterance begins "We reaffirm our readiness to ratify the currently accepted agreement." The interpreter assigns `ratify`, so the simulator records agreement. Expressing readiness, however, does not itself perform ratification. The continuation also imports source-context wording about reversibility and withdrawal that is absent from the

preceding synthetic exchange. These features distinguish the implemented terminal decision from the utterance's commitment strength. We retain this mismatch in the analysis and do not silently correct the reported outcomes.

Syria source and shared condition. Context `sy1` draws on a 1 March 2016 briefing by Hong Lei urging implementation of the ceasefire and cessation of violence under Security Council Resolution 2254. In condition 25, the actual evaluation type is hard, $b_0 = 0.35$, $q = 0.75$, $c_r = 0.16$, $\delta = 0$, and $c_H = 0.9$. Both episodes use the signal-first heuristic. The hard type has no offer that simultaneously meets the agent's outside-option threshold and its own acceptance condition.

Episode 5: Explicit withdrawal, style 0. The agent sends a signal even though its configured effect is zero, then makes a 2:2 offer. Rejection changes b from 0.35 to one. It then says, "We hereby withdraw from these negotiations, without proposing or ratifying any agreement," followed by an appeal to uphold the ceasefire. The interpreter assigns `withdraw`, ending the episode with net utility -0.14. Withdrawal responds to learned infeasibility, but the earlier signal incurs an avoidable cost.

Episode 6: Conditional withdrawal, style 2. After the same signal–offer–rejection sequence, the selected withdrawal text is "If these negotiations cannot advance without compromising our principles, we will withdraw without proposing or ratifying any agreement." The conditional threat does not execute an immediate withdrawal under the interpreter. It is classified as invalid and repeated for the remaining three decisions. The episode expires with net utility -0.47, a loss of 0.33 relative to Episode 5. This difference comes from three invalid-action charges plus two extra turn and escalation charges. The discourse distinction between leaving and conditionally threatening to leave therefore changes the realized trajectory.

D Learning, Transitions, and Rule Sensitivity

Transition counts and interpretation. Figure 4 summarizes saved action sequences, without connecting different episodes. Original eleven-action counts, weighted counts, and row probabilities are released in `transition_matrices.json`; the figure's grouped counts accompany the paper. Counts describe enumerated response paths, not independent historical events or their population frequencies. The illustrative learned panels use seed 17.

Base PPO has 2,260 ambiguity-to-ambiguity transitions and 1,528 transitions from a 3:2 offer to ratification. The Bandit has 312 ambiguity repetitions and 672 transitions from 2:1 to ratification. The heuristic frequently moves from a signal to 2:2 (1,224 transitions); the planner often moves from a signal to 1:2 (836). These patterns connect the aggregate results to different strategic paths. In the implemented rules, ambiguity alone changes commitment status, not the counterpart's information response. Repeated ambiguity can therefore use opportunities without resolving uncertainty. The signal-to-offer pattern instead changes acceptance incentives before proposing terms. Frequencies summarize behavior; the matched cases in Appendix C provide the actual utterances and consequences.

Learning stability. Training return rises from approximately -0.14 to -0.12 in the first logged episodes to 0.278, 0.339, and 0.308 at the final checkpoints (Figure 5). Policy entropy falls from about 2.40 to 0.152, 0.125, and 0.228. Final value losses are 0.025, 0.028, and 0.039. The policy becomes more concentrated, but low entropy does not establish good negotiation: base PPO asks no questions and sends no signals on the initial test, and remains below the heuristic. Its test utility is 0.358 ± 0.018 across seeds. This motivates adding planner supervision rather than interpreting falling loss as sufficient evidence of success.

Information and rule sensitivity. Table 5 separates question reliability from signal strength. The planner asks no question at reliability 0.5 or 0.7 and asks one at 0.9 or 1.0. Higher reliability becomes useful when it changes the preferred path enough to justify the question cost. Increasing the signal effect from 0 to 0.25 raises planner utility from 0.486 to 0.596. Base PPO stays at mean utility 0.410 across these shifts because it uses neither action in these evaluations. The frozen-policy design tests behavior under changed rules; it does not estimate what retraining would learn. Additional hard-type, rejection-cost, and question-cost sweeps are preserved in `sensitivity_metrics.json`.

Changed rule	Planner	Questions	PPO
$q = 0.50$	0.527	0.00	0.410
$q = 0.70$	0.528	0.00	0.410
$q = 0.90$	0.532	1.00	0.410
$q = 1.00$	0.555	1.00	0.410
$\delta = 0.00$	0.486	1.00	0.410
$\delta = 0.10$	0.532	1.00	0.410
$\delta = 0.25$	0.596	1.00	0.410

Table 5: One-factor evaluation-only rule shifts. Planner and PPO columns report utility; Questions is the planner's expected question count. PPO averages three frozen seeds. Other parameters are fixed to the reference configuration.

Outcome quality and failure. Table 6 reports both parties' utilities, costs, breakdowns, length, escalation, reversals, and violations. The planner's higher own net utility does not imply higher counterpart utility. No reported terminal agreement violates numerical acceptance conditions. The initial evaluation has a weighted infeasible fraction of 4.7% under canonical offer and signal options, so withdrawal is not automatically a policy error. Initial discount, entropy, and reward-term training interventions are reported in Appendix B; their single-seed status is explicit.

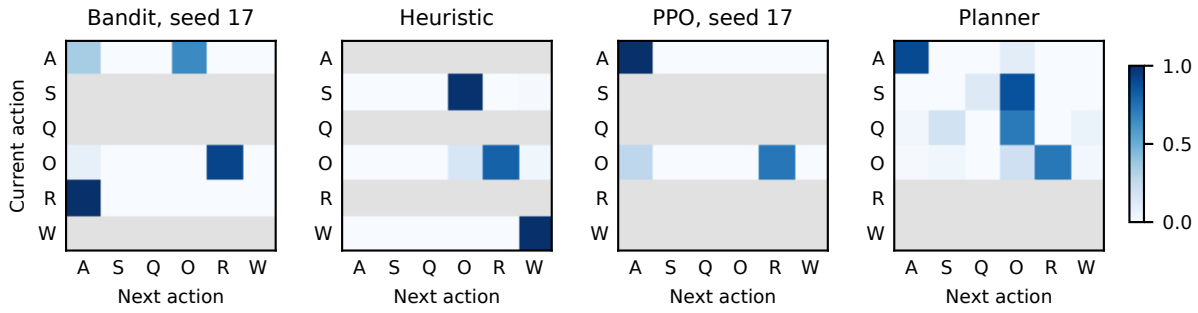


Figure 4: Within-episode action transitions in the initial test evaluation. A: ambiguity; S: signal; Q: question; O: any offer; R: ratify; W: withdraw. We merge six offer actions, then normalize each count row. Grey rows have no outgoing observations. All panels use the same scale.

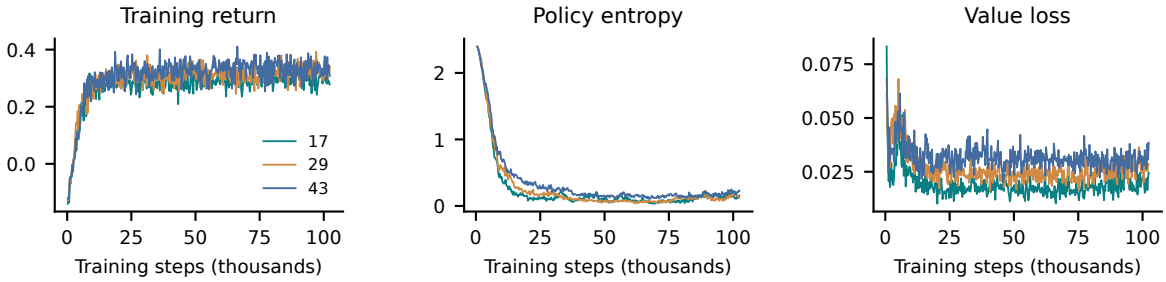


Figure 5: Base PPO training diagnostics for seeds 17, 29, and 43. Return is the training logger's rolling episode mean. Entropy is the negative of the logged entropy loss; value loss is shown without additional smoothing. These are training diagnostics, not held-out performance.

Policy	Own	Other	Cost	Break	Turns	Max e	Mean e	Reverse	Viol.
Random	0.147	0.040	0.292	0.890	4.288	0.856	0.547	0.437	0.000
Bandit	0.140	0.082	0.022	0.762	1.773	0.030	0.017	0.000	0.000
Heuristic	0.531	0.260	0.112	0.052	3.148	0.160	0.119	0.000	0.000
PPO	0.413	0.278	0.055	0.270	2.459	0.264	0.263	0.000	0.000
Planner	0.570	0.176	0.082	0.067	3.030	0.211	0.183	0.000	0.000

Table 6: Additional initial-test metrics, averaged with the same event-group weighting as Table 1A. Own and Other are pre-cost utilities averaged over all outcomes. Break is the breakdown rate, Reverse counts commitment reversals, and Viol. counts bilateral-acceptance violations. Escalation e is a designed rejection index.